

# Comparación de Rendimiento: Procesador Baseline vs Descompresor

Generado: 2025-11-24 02:41:53  
Proyecto de Aplicación - II Semestre 2025  
Jose Antonio Espinoza Chaves | 2019083698

## DESCRIPCIÓN DEL SISTEMA

Este reporte compara el rendimiento de dos implementaciones de procesador ARM32:

- Procesador Baseline
  - Pipeline estándar de 5 etapas sin compresión
  - Instrucciones de 32 bits
- Procesador con Descompresor
  - Pipeline de 5 etapas con módulo de descompresión
  - Instrucciones comprimidas (tokens de 8 bits)
  - Descompresión en tiempo real por hardware

## RESULTADOS CLAVE

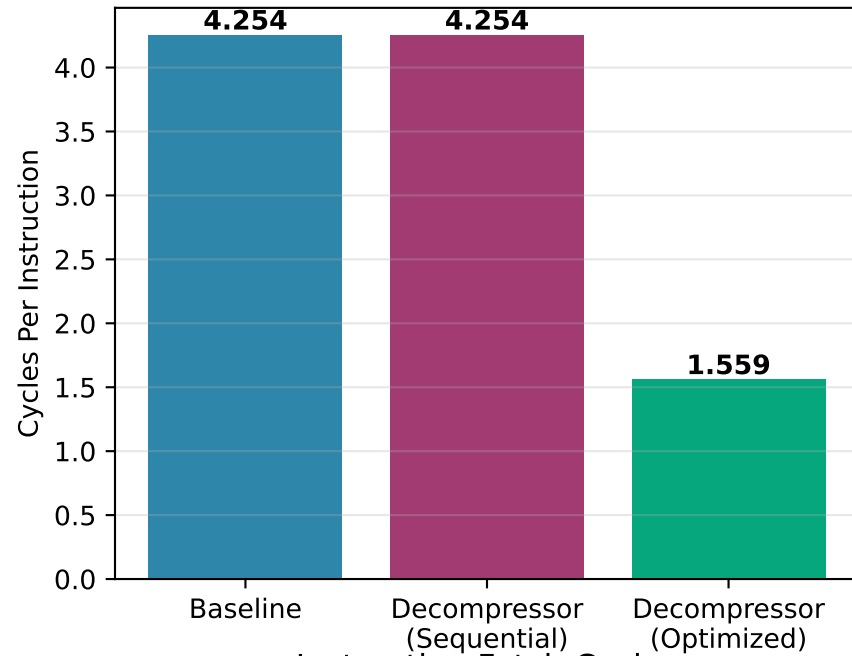
|                        |       |
|------------------------|-------|
| Compresión:            | 69.9% |
| Reducción Ancho Banda: | 56.2% |
| CPI Baseline:          | 4.254 |
| CPI Optimizado:        | 1.559 |
| Mejora en CPI:         | 63.3% |

## MÉTRICAS COMPARADAS

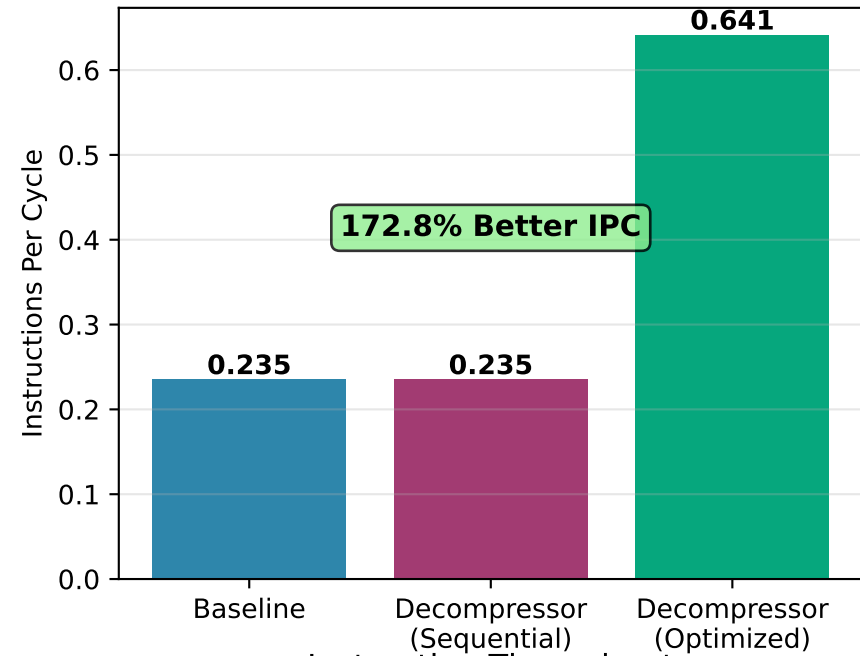
- ✓ Ciclos de Ejecución
  - ✓ Instrucciones Ejecutadas
  - ✓ CPI / IPC
  - ✓ Utilización del Pipeline
  - ✓ Ancho de Banda de Memoria
  - ✓ Throughput de Instrucciones
  - ✓ Speedup en Fetch de Instrucciones
- Las métricas optimizadas asumen fetch paralelo de tokens (4 tokens por fetch de 32 bits)*

# Optimized Performance

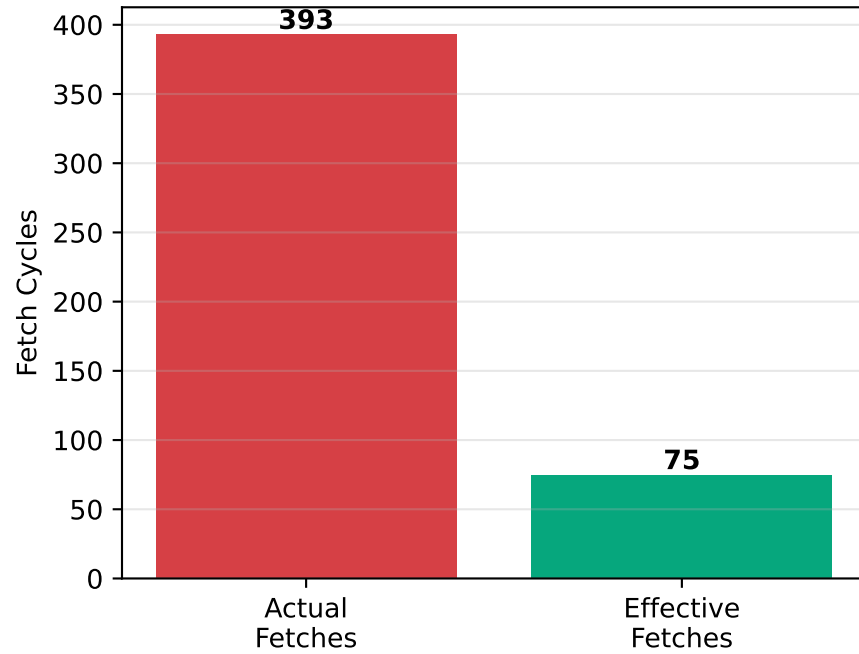
## CPI Comparison



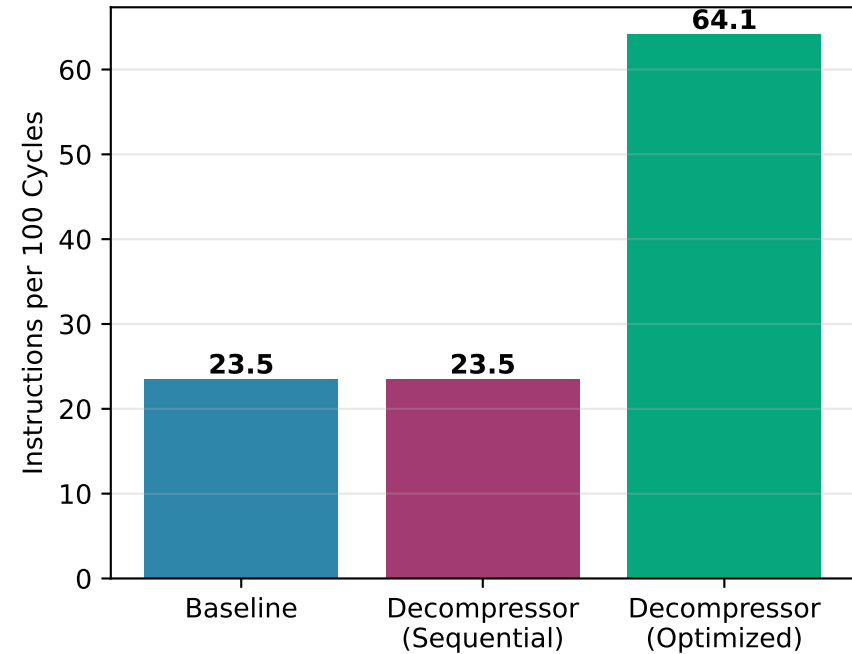
## IPC Comparison



## Instruction Fetch Cycles



## Instruction Throughput



# Detailed Metrics Comparison: Baseline vs Optimized Decompressor

| Metric                       | Baseline | Optimized | Difference | % Change |
|------------------------------|----------|-----------|------------|----------|
| cpi                          | 4.2542   | 1.5593    | -2.6949    | -63.35%  |
| ipc                          | 0.2351   | 0.6413    | +0.4062    | +172.83% |
| pipeline_utilization         | 23.5060  | 23.5060   | +0.0000    | 0.00%    |
| instruction_throughput       | 23.5060  | 64.1304   | +40.6244   | +172.83% |
| total_cycles                 | 502      | 502       | +0         | 0.00%    |
| total_instructions           | 118      | 118       | +0         | 0.00%    |
| execution_latency            | 255      | 255       | +0         | 0.00%    |
| instruction_memory_bandwidth | 25.0518  | 10.9801   | -14.0717   | -56.17%  |
| memory_bandwidth             | 112      | 112       | +0         | 0.00%    |
| instruction_fetches          | 393      | 393       | +0         | 0.00%    |
| effective_fetch_cycles       | 0        | 75        | +75        | ∞        |
| fetch_speedup                | 0        | 5         | +5         | ∞        |
| alu_operations               | 92.0000  | 93.0000   | +1.0000    | +1.09%   |
| memory_accesses              | 56.0000  | 56.0000   | +0.0000    | 0.00%    |
| compression_ratio            | 0        | 70        | +70        | ∞        |
| decompressed_instructions    | 0        | 253       | +253       | ∞        |
| decompressed_nops            | 0        | 141       | +141       | ∞        |
| instruction_fetch_savings    | 0        | 70        | +70        | ∞        |
| memory_access_rate           | 11.1554  | 11.1554   | +0.0000    | 0.00%    |
| memory_efficiency            | 0        | 0         | +0         | ∞        |
| memory_reads                 | 28.0000  | 28.0000   | +0.0000    | 0.00%    |
| memory_writes                | 28.0000  | 28.0000   | +0.0000    | 0.00%    |
| nops                         | 244      | 244       | +0         | 0.00%    |
| register_writes              | 29.0000  | 29.0000   | +0.0000    | 0.00%    |
| total_processed              | 0        | 362       | +362       | ∞        |

Green = Optimized is Better | Red = Optimized is Worse | Gray = Same or Neutral